

Presented by:
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ALX DATA SCIENCE:
MACHINE LEARNING PROJECT

MOVIE RECOMMENDATION SYSTEM

BUILDING AND EVALUATING COLLABORATIVE FILTERING MODELS WITH MOVIELENS



PROJECT OVERVIEW

01

Dataset

MovieLens dataset **containing 10+ million user ratings.**

04

Evaluation

RMSE (Root Mean Squared Error) — lower values indicate better performance.

02

Competition

Kaggle recommendation-system competition completed as part of the **ALX Data Science programme.**

05

Approach

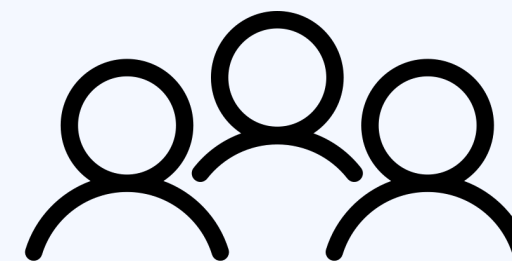
Compared **Global Mean, User + Movie Bias, SVD, hyperparameter tuning, and ensemble modelling** to identify the best-performing model.

03

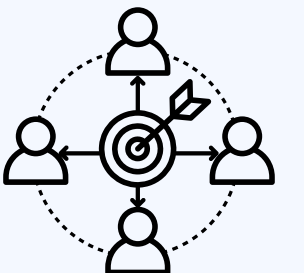
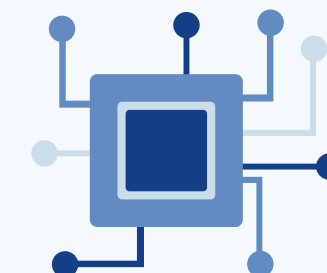
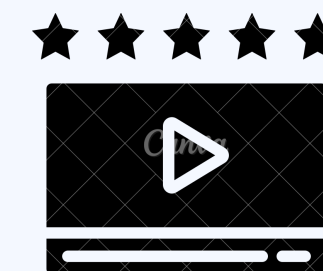
Objective

Predict the rating a user would give to a **movie they have not yet rated.**

USERS



» MOVIE RATINGS » MACHINE LEARNING MODEL » PREDICTED RATINGS



DATASET OVERVIEW

Training Data

- **10,000,038 ratings**
- Columns: **userId**, **movieId**, **rating**, **timestamp**
- Ratings range from **0.5 to 5.0**

Test Data

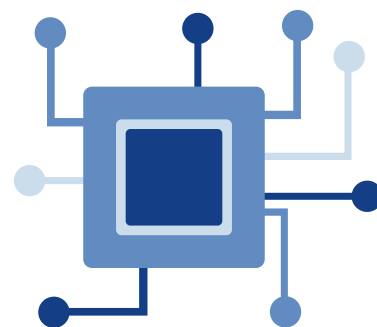
- 5,000,019 user-movie pairs
- Contains **userId** and **movieId**
- Ratings are not provided — these are what our model must predict

Additional Movie Information

- Movie genres
- Tags and tag-genome information
- IMDb metadata
- Movie ID mappings



TRAIN
10M+
ratings



Machine
Learning
model



TEST
5M+ user-
movie pairs



Predicted
Ratings

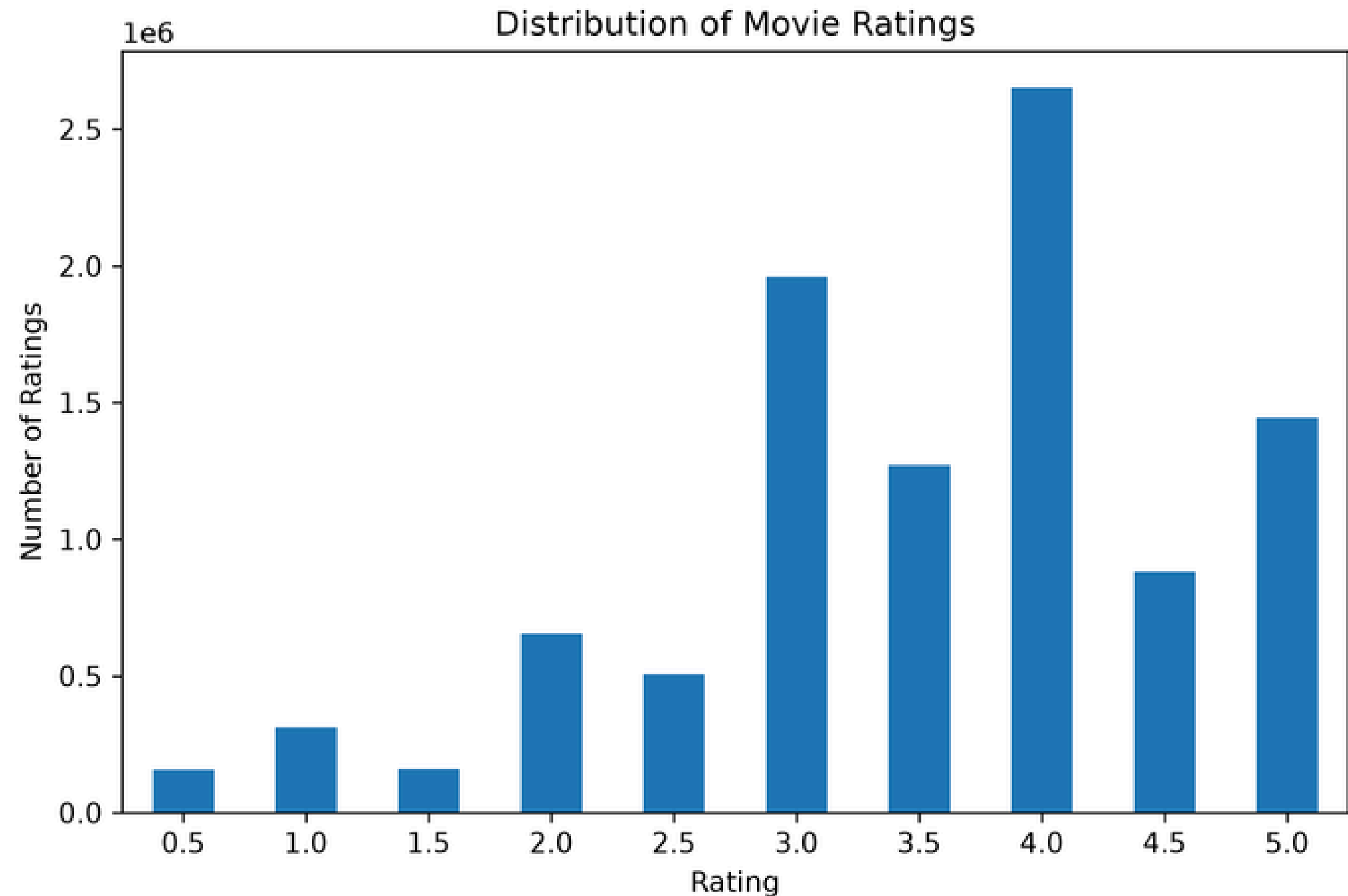
EXPLORATORY DATA ANALYSIS

01 Rating Distribution:

- 0.5-5.0 half star scale
- Uneven distribution accross all ratings

02 Key Findings:

- Mean rating: 3.53/5
- Uneven user/movie activity
- Global mean provides a useful baseline



BASELINE MODEL — GLOBAL MEAN

What is the Global Mean?

The average rating calculated from the training dataset.

HOW IT WORKS

1

Calculates the overall average rating.

Training Ratings

2

Predicts this same value for every user–movie pair.

Calculate Mean
3.5335

3

Provides a simple benchmark for evaluating more advanced models.

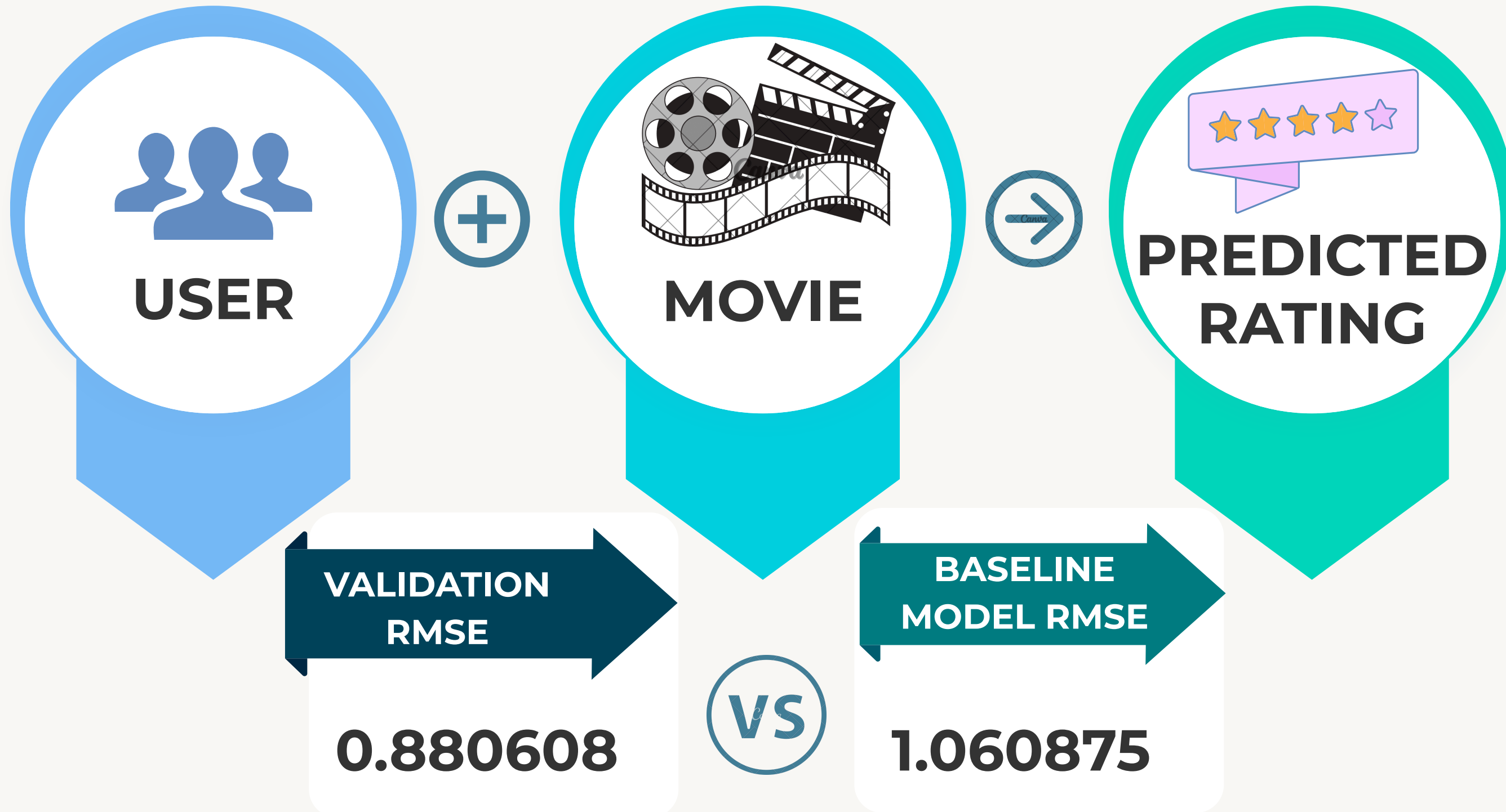
Predict **3.5335** for every user–movie pair



Result

Validation
RMSE:
1.060875

USER + MOVIE BIAS MODEL



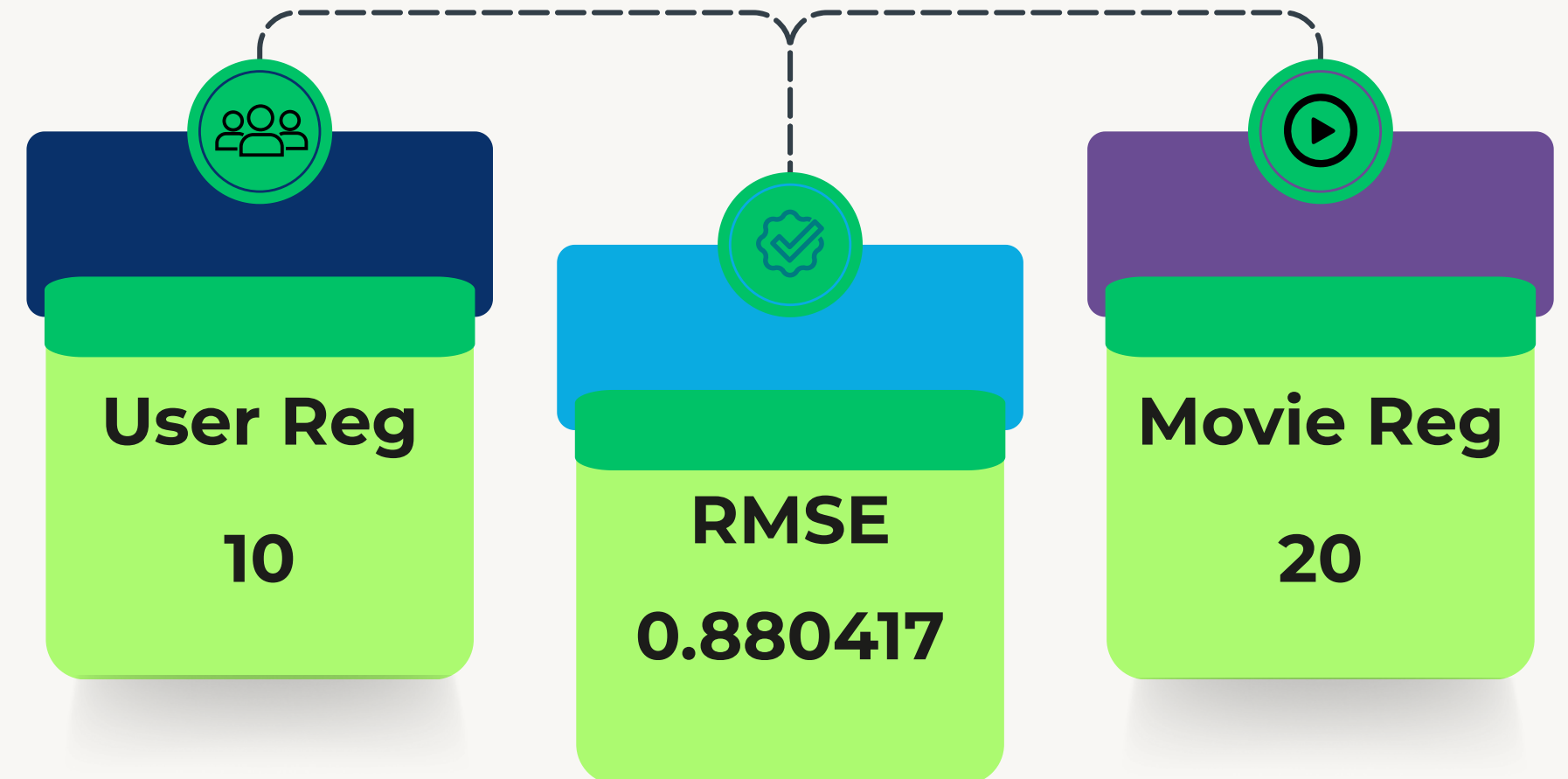
Key note:

Capturing user and movie tendencies reduced RMSE by 17% compared with the Global Mean baseline model.

BIAS MODEL — TUNING

User Reg	Movie Reg	RMSE
10	10	0.880078
10	20	0.880417
20	20	0.880734

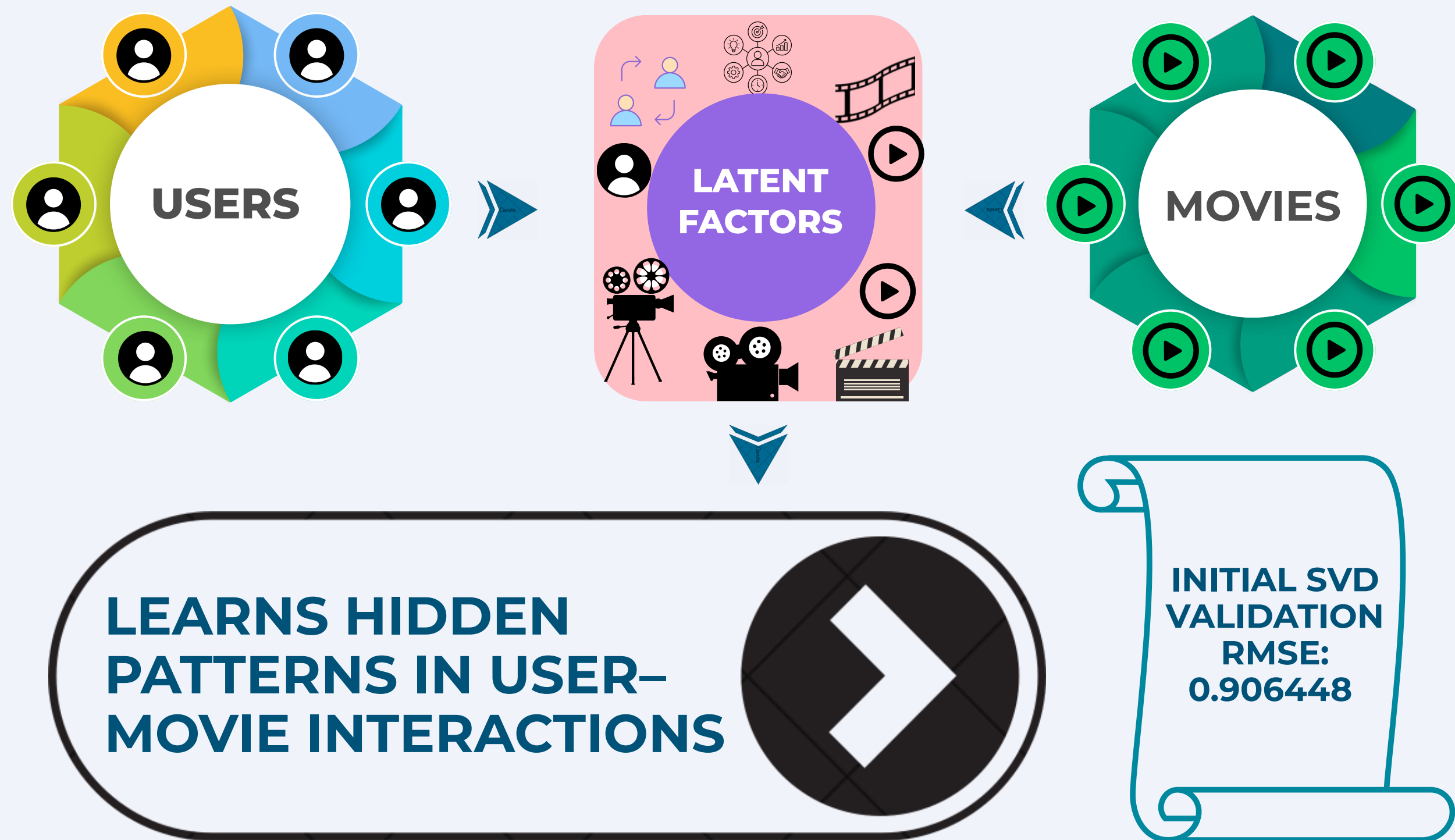
BEST CONFIGURATION 



Key note:

Tuning reduced RMSE and produced the strongest individual model.

SVD — MATRIX FACTORIZATION



Key note:

Improved over the Global Mean, but did not outperform the tuned Bias model.

SVD — HYPER-PARAMETER TUNING

Parameter explored

Latent factors

0.906448

50 factors, 20 epochs

Epochs

0.900959

20 factors, 20 epochs

Regularization

0.898068

20 factors, 20 epochs, reg: 0.05

BEST SVD CONFIGURATION

Latent factors:
20

Epochs:
20

Reg:
0.05

Initial SVD
Validation RMSE:
0.898068

Key note:

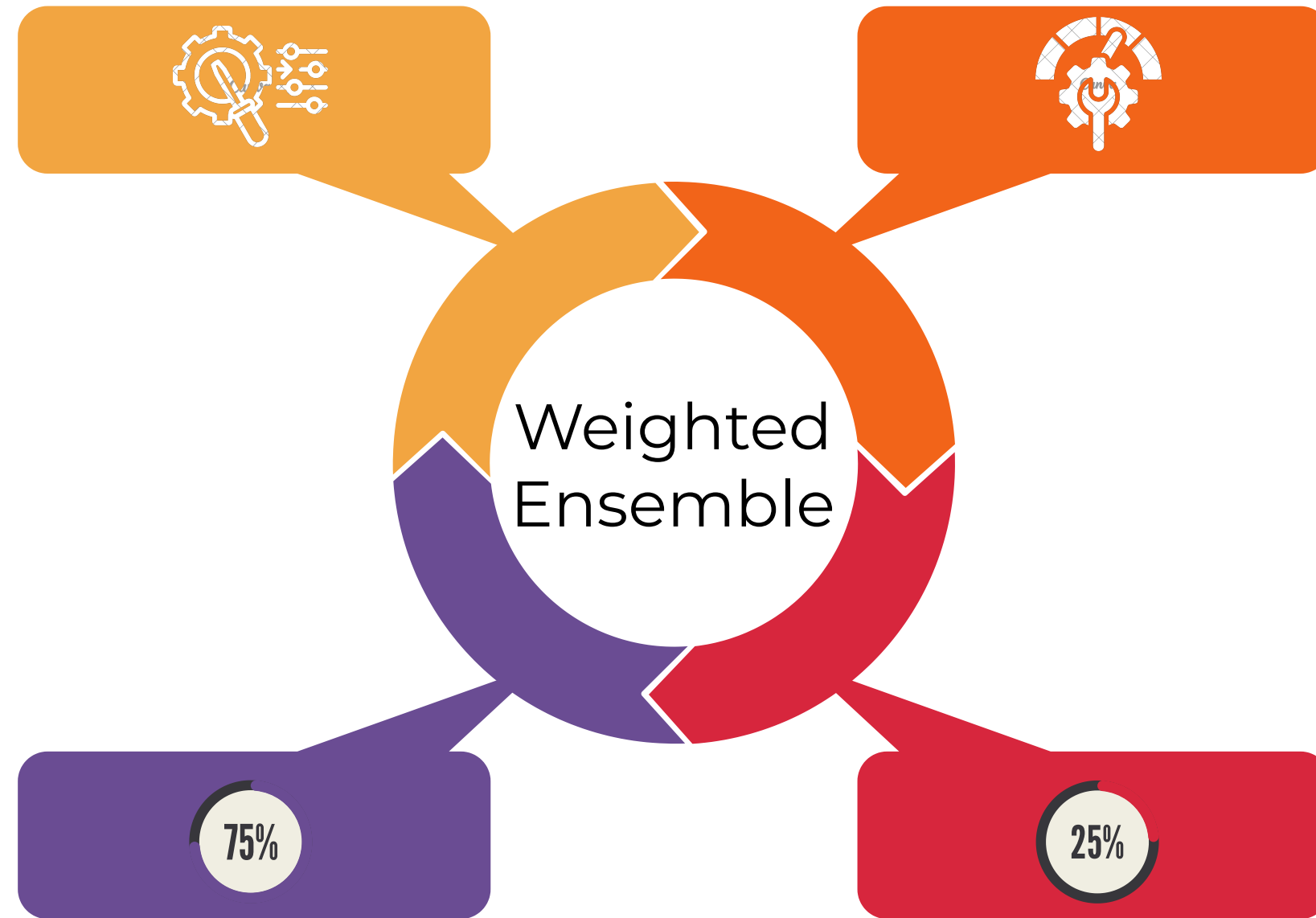
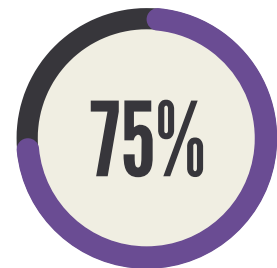
Hyper-parameter tuning improved SVD performance.

MODEL ENSEMBLE

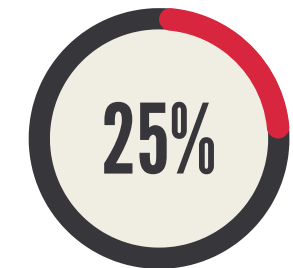
Tuned User + Movie Bias
RMSE: 0.880417

Tuned SVD
RMSE: 0.898068

User + Movie Bias



SVD

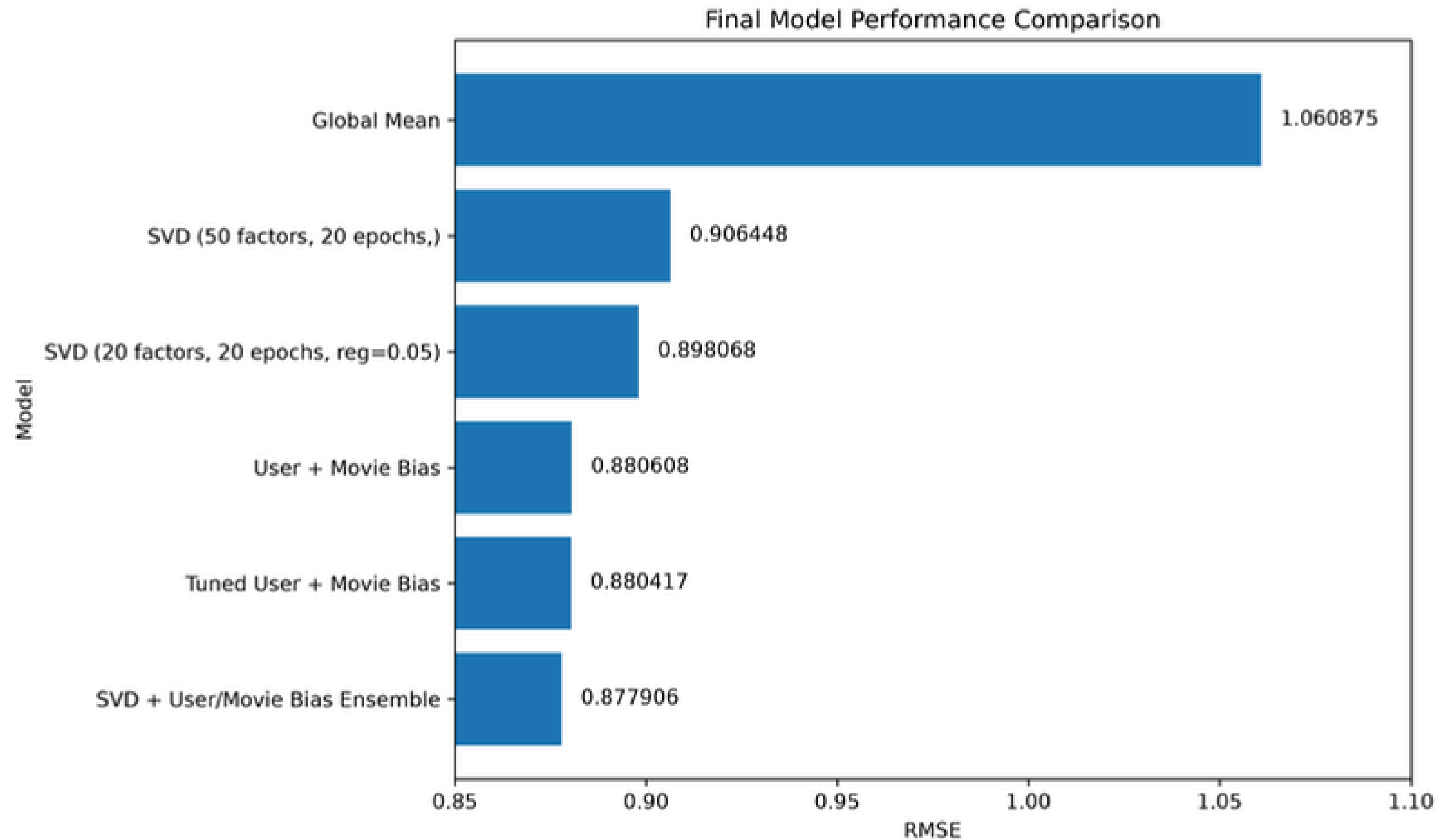


Validation RMSE: 0.877906

Key note:

The ensemble achieved the lowest validation RMSE among all tested models.

MODEL COMPARISON & FINAL SELECTION



Key note:

Lower RMSE = Better Performance
Best validation RMSE: 0.877906

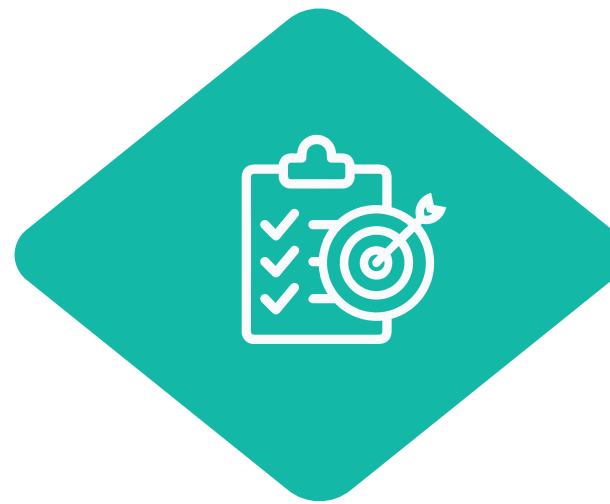
FINAL PREDICTIONS & SUBMISSION



01

SVD + Bias Ensemble

Validation RMSE:
0.877906



02

Test Set Predictions

5,000,019
User–Movie Pairs

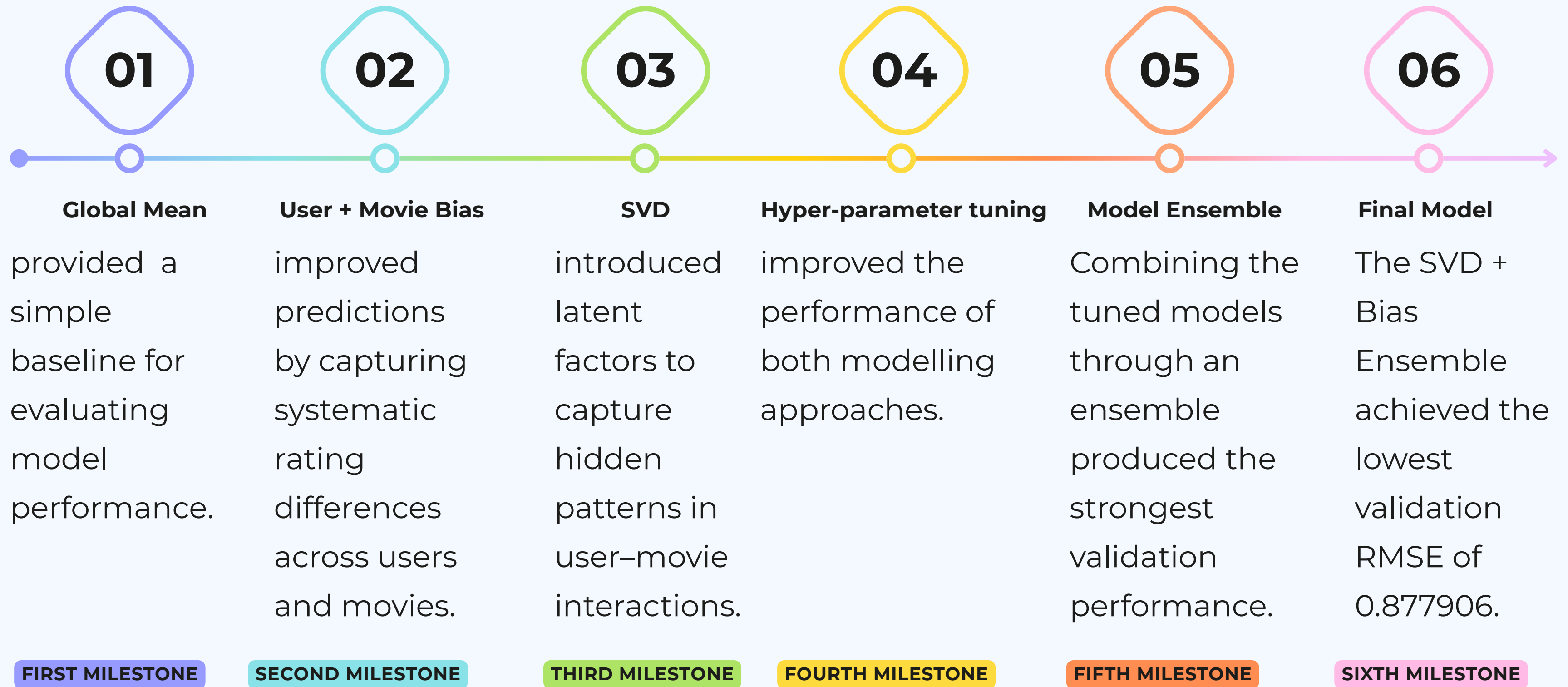


03

Submission file

[final_submission_corrected.csv](#)
Final predicted ratings
submitted for evaluation

CONCLUSION



Thank You!

Let's keep in touch.
Reach out for any questions or
clarifications.

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